TRL Rent Management – Guide

Features · Running and operating the app · Essential information. Version deployed 13 Oct 2026 (TRL 7295654, TRL_API e47a63e).

1. Features (what each menu does)

Menu	What you do there
Dashboard	Rent due vs collected, occupancy and arrears at a glance.
Tenants	Add / edit tenants (individual or company). A tenant can't be deleted while they have an active lease, an open balance or a deposit held.
Buildings & Units	Buildings → floors → units (type, base rent, status). Deleting is refused while something is still in use.
Unit Occupancy	Which units are occupied or vacant.
Lease Management	Create a lease (tenant, unit, rent, start date, months). Renew, end (move-out) or cancel. One active lease per unit. A tenant's unit and rent always come from their lease.
Rent Collection	Generate monthly invoices; record payments (Cash, Bank Transfer, Cheque, Online) with optional discount or late-fee waiver; charge late fees; extra charges; change due dates; cancel an invoice (with a reason). Print a receipt after paying.
Rent History	All invoices and their history. **Reverse** a wrong payment here (payments are never edited or deleted – a reversal adds a matching negative record). Print invoice.
Payment Records	All payments in a date range, receipt links, CSV export.
Security Deposits	Agreed deposit per tenancy, record received amounts, corrections, deposit receipts. Kept separate from rent income.
Move-out Settlements	After a lease has ended: dues vs deposit, deductions, apply deposit, refund or balance due. Printable settlement.
Reminders	Unpaid / overdue invoices to follow up.
Maintenance	Maintenance jobs, cost, status; optionally bill the cost to a tenant.
Reports	**Arrears Ageing** (who owes, how long) · **Collections** (cash received by date) · **Tenant Statement** (running balance) · **Billing vs Collection** (per month) · **Occupancy / Rent Roll** · **Maintenance Cost**. All can be printed or exported (CSV).
Company Profile	Phone, e-mail, NTN, address, logo – printed as letterhead on receipts, invoices and reports.
Late Fee Settings	Grace days, fee per day, maximum (default 5 days, 500/day, max 2 × invoice rent). Applies to invoices generated after the change.
Change Password	Account menu (top right). Logs out other devices.

Rules to know: only Admin users can log in · each company's data is in its own database · dates use Pakistan time · payments are corrected only by reversal · the database itself blocks invalid money data.

Typical monthly flow

1. New tenant: **Tenants** → add → **Lease Management** → create lease → **Security Deposits** → record deposit.
2. Each month: **Rent Collection** → Generate Invoices → record payments as they come in → print receipts.
3. Follow up: **Reminders** / **Arrears Ageing**.
4. Move-out: **Lease Management** → end lease → **Move-out Settlements** → finalize → print.

2. Running and operating the app

Run on your PC (development)

1. SQL Server Express `.\SQLEXPRESS` must be running (databases `TRL_DB` and `TRL_Catalog`).
2. API: open `D:\TRL\TRL_API` in Visual Studio and run the **https** profile (or `dotnet run --project TRL_API --launch-profile https`) → `https://localhost:7295`.
3. Frontend: in `D:\TRL\TRL` run `npm install` (first time) then `npm start` → open the address it prints (e.g. `http://localhost:3000`).
4. Log in with company code `TRL` and your Admin user.

Tests: frontend `npm test` · API end-to-end `node e2e/run.mjs` (in TRL_API; uses temporary databases only) · API unit tests `dotnet test TRL_API.Tests`.

Deploy a new version (order matters)

1. **Backup** the production database (below).
2. **Database change** (only if the release has one): run its script in MonsterASP WebMSSQL on db71356 and check the result row. WebMSSQL scripts must start with `SET NOCOUNT ON`; "Caution: Changing any part of an object name..." lines are harmless.
3. **API**: Visual Studio → TRL_API → Publish → profile `site95547-WebDeploy`.
4. **Frontend**: `npm run build` (do **not** set `VITE_API_BASE_URL`: the app calls `/api` on its own site and Netlify forwards it to the API, so logins work on every browser and phone), then Netlify → Deploys → drag the `build` folder (or push to GitHub if Netlify builds from it).
5. **Check**: `powershell -ExecutionPolicy Bypass -File D:\TRL\deploy_ready\phase6\post_deploy_check.ps1` → must end with **ALL CHECKS PASSED**.

Undo: Netlify → previous deploy → Publish; API → publish the previous build; database → its script in `Database/rollback` (WebMSSQL: the `_v3_ROLLBACK` files).

Backups (most important)

- **Every week** and before every change: MonsterASP → Databases → db71356 → Backups → **Create BAK file** → Download. Keep a copy off the PC (cloud).
- Keep the last 8 weekly + 12 monthly backups.
- Restore only after data loss: panel → Backups → restore. Everything recorded after that backup must be re-entered.

Monitoring and logs

- Health: `https://rent-trlapi.runasp.net/api/health` → `{"status":"ok"}`. Set up UptimeRobot (free) on it, every 5 minutes, alert by e-mail.
- Logs: MonsterASP File Manager → site95547 → `logs/trl-<date>.log` (failed logins, errors; kept 30 days). An error shown to users has a "Reference" code – search it in the log.

Add a new company (client)

```
dotnet run --project TRL_Tools -- provision --code CODE --name "Company Name" --connection-key
KEY --admin-user USER --create-database
```

Then add the setting `ClientConnections__KEY` (its database connection) on the API host. Details: `TRL_API/Database/README.md`.

If something goes wrong

Problem	Cause / fix
"This company's database needs an update"	A database migration wasn't applied. Run it (WebMSSQL in production, `TRL_Tools migrate --client TRL` locally).
Health shows 503	Same as above, or the database can't be reached. Check the log file.
Locked out after wrong passwords	Wait 15 minutes and try again.
Freshly built API won't start on your PC	Windows Smart App Control blocked the new DLL – rebuild, or run it from another folder.
Page doesn't load at all	Check Netlify (frontend) and the health link (API).

3. Essential information

Item	Value
Website (frontend)	`https://voluble-lokum-rent.netlify.app` (Netlify)
API	`https://rent-trlapi.runasp.net/api` (MonsterASP site95547)
Production database	db71356 (SQL Server 2025) – only through MonsterASP **WebMSSQL**; catalog db72317
Company code	`TRL`
Source code	`D:\TRL\TRL` (frontend, React/Vite) · `D:\TRL\TRL_API` (API .NET 8, database scripts, tools, tests)
Deployment packages	`D:\TRL\deploy_ready\phase6` (current) · `deploy_ready\api` (previous API, for undo)
Production SQL scripts	`D:\TRL\prod_import` – use the `_v3_` files; `verify_13_READONLY.sql` checks the database
Documentation	`TRL_API/OPERATIONS.md` (operations) · `TRL_API/Database/README.md` (database) · `deploy_ready/phase6/DEPLOY_PHASE6.txt`
Local databases	`TRL_DB`, `TRL_Catalog` on `.\SQLEXPRESS`

Still to do

- In about a week (after a new backup): run `11W_v3_db71356_drop_tenant_unit_copy_WEBMSSQL.sql` in WebMSSQL – removes 4 unused old columns. Optional; nothing breaks without it.
- Set up UptimeRobot on the health link; check once that the log file is written.
- Before real customers: change the Admin password, and delete `prod_import\users.sql` and `trl_prod_import.sql` (they contain the old password hash).

> **Security:** never share passwords, connection strings, backups or the `prod_import` folder. Passwords are not stored in this document.

Possible later improvements

User roles (manager, accountant, view-only) · audit log · stronger hosting (Azure) with staging and automatic deployment · two-factor login.